SUPPLEMENTARY RESULTS

Paper ID 315

1. PIDNet-S Pixel-Level Evaluation

Results on the 84-image test split. Pixel metrics are pooled at original annotation resolution.

Metric	512 × 384	320 × 240
Test images (n)	84	84
Mean IoU (%)	70.14	68.01
Not-floor IoU (%)	87.42	86.37
Floor IoU (%)	52.86	49.64
Pixel accuracy (%)	88.97	87.99
Floor Dice (%)	69.16	66.34

Table 1. Segmentation results at the two evaluated input resolutions (%).

Table 1 compares the checkpoint input size (512 × 384) with a lower-resolution input (320 × 240). At 512 × 384, mean IoU is 70.14% and pixel accuracy is 88.97%; reducing the input resolution lowers mean IoU to 68.01% and floor IoU from 52.86% to 49.64%.

Input	TN	FP	FN	TP
512 × 384	46,712,116	1,383,320	5,341,012	7,540,430
320 × 240	46,431,174	1,664,262	5,661,202	7,220,240

Table 2. Pooled pixel confusion counts; rows are ground truth, columns are predictions.

Table 2 gives the pixel counts behind the scores. False negatives are floor pixels missed as not-floor; false positives are not-floor pixels incorrectly labeled as floor. The larger floor false-negative share explains why floor IoU is the weaker class metric.

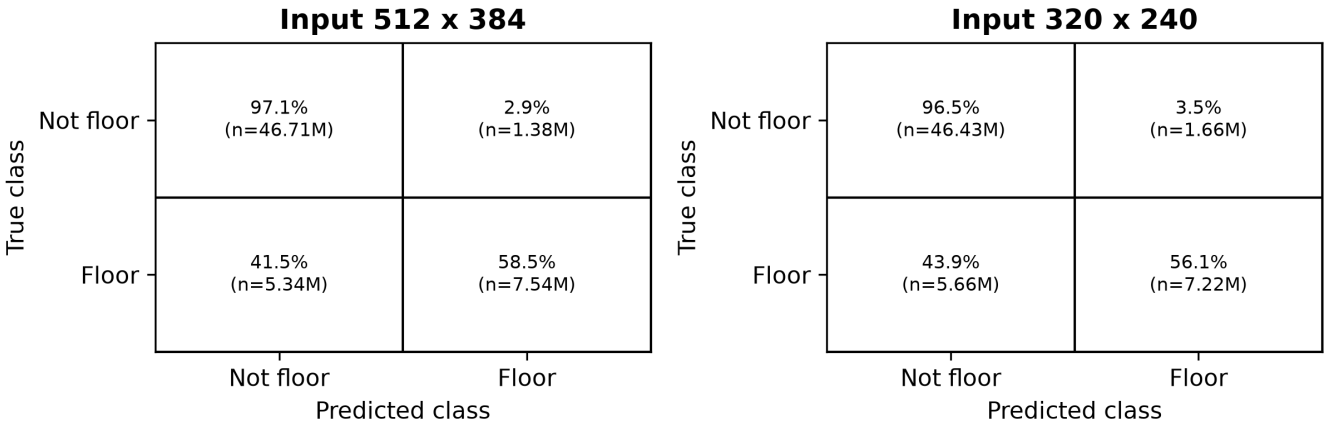


Figure 1. Pixel-level confusion matrices. Class 0 = not-floor; class 1 = floor.

At the checkpoint input size, the model achieved 70.14% mIoU and 88.97% pixel accuracy. At 320 × 240, mIoU decreased by 2.13 percentage points and floor IoU by 3.22 points. In both confusion matrices, floor pixels were more often missed as not-floor (41.5% and 43.9% of ground-truth floor pixels) than not-floor pixels were labeled as floor (2.9% and 3.5%).

The evaluated checkpoint is PIDNet-S. Exact image-hash checks found no identical images across train, validation, and test splits; this does not establish scene- or session-level independence. The 320 × 240 evaluation uses Pillow bilinear resizing, whereas the live implementation uses OpenCV area resizing.

2. Qualitative Segmentation Results

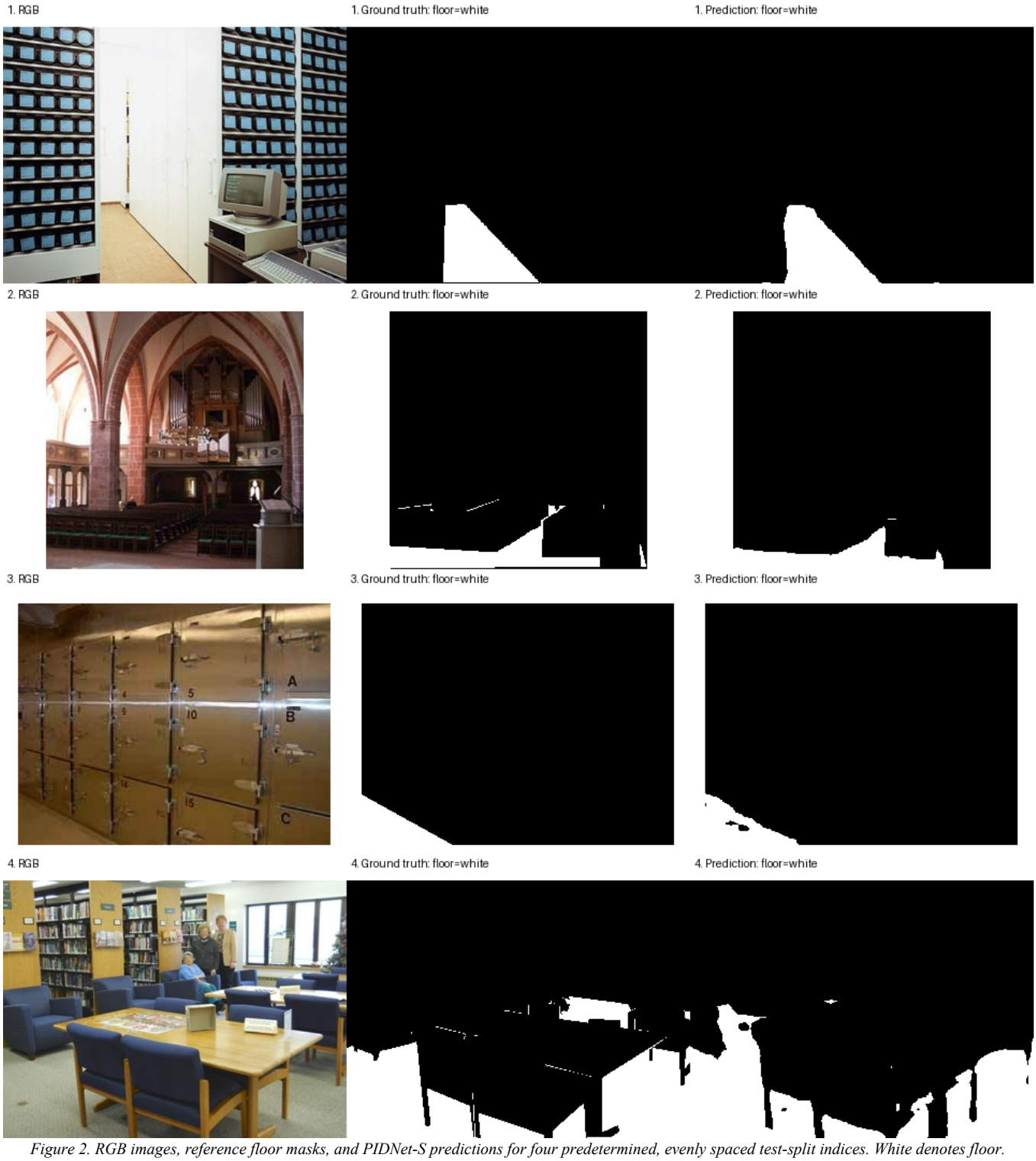


Figure 2. RGB images, reference floor masks, and PIDNet-S predictions for four predetermined, evenly spaced test-split indices. White denotes floor.

The examples show that the predicted masks follow broad floor regions, while local differences remain around object boundaries and visually complex scenes. These selected examples illustrate output appearance; the pooled pixel metrics above summarize the full test split.

3. Single-Frame Results on the Two Supplied Scene Photos

Area shares are model predictions after resizing the supplied portrait photos to a 640 × 480 runtime frame.

Photo	Floor (%)	Not-floor (%)	ROI floor (%)	Min. cell floor (%)	Single-frame trigger
Photo 1	63.74	36.26	100.00	100.00	No
Photo 2	72.33	27.67	100.00	100.00	No

Table 3. Predicted floor/non-floor shares and single-frame screening outputs.

The predicted whole-frame floor share is 63.74% in Photo 1 and 72.33% in Photo 2. In both frames, all evaluated bumper-ROI cells are classified as floor, so neither single-frame screening rule triggers. This describes the output of the current segmentation and threshold rule on these two photos only.



Figure 3. Supplied scene photos, predicted floor masks, and runtime overlays. Green = predicted floor; orange = bumper ROI.

The model has floor and not-floor classes only. Not-floor includes furniture, walls, and background; it is not an obstacle label. No pixel-level ground truth or temporal confirmation is available for these two single frames. “No trigger” reports only the configured screening rule output and does not establish obstacle-detection accuracy or safe navigation.

4. Robot Trial Results Reported in the Main Paper

Sections 4–6 reproduce aggregate values stated in the supplied main-paper PDF. The PDF did not include row-level query labels, advisor rating sheets, or per-trial logs, so these summaries are transcribed rather than independently recalculated.

The vision-based reactive floor-following system using fine-tuned PIDNet-L was evaluated in campus corridors under varying lighting conditions. The camera perception loop achieved an average inference frame rate of 19.8 Hz on the Mini PC. Trials were conducted on a 40-meter hallway route and were counted as operationally successful when the robot completed the route without a reported obstacle contact. This experiment evaluates movement behavior rather than collision-safe autonomous navigation.

The conclusion reports an average response time of 11.26 s. The paper does not report numerical PID gains or physical operating speed in m/s, and it does not describe the two incomplete trials individually.

The live-trial result is reported for PIDNet-L. The separate 84-image pixel-level benchmark in Sections 1–2 evaluates the available PIDNet-S checkpoint; these are distinct model evaluations.



Figure 4. Left panel: HIRA robot used in the experimental setup. Right panel: Real-time floor segmentation and reactive PID steering control in a university hallway.

The following explanation is transcribed from the main paper; its Fig. 6 is reproduced here as Figure 4.

As illustrated in Fig. 6, the system segments the traversable floor area and produces steering commands. The trapezoidal ROI and temporal confirmation counter (10 frames) reduced transient false negatives in the reported setup. Over 50 reactive floor-following trials, 48 trials completed without a reported obstacle contact, corresponding to a 96.0% operational success rate. Because the study used a single RGB camera and did not measure distance, near-collision events, segmentation IoU, or confidence intervals, this result should not be interpreted as a safety guarantee.

5. Retrieval Benchmark and Ablation Results Reported in the Main Paper

On the 50-query institutional benchmark, the full system reaches 90.0% Recall@1, 94.0% Recall@3, and 0.920 MRR. With title weighting, intent boosting, and title inheritance retained, enabling graph traversal adds 4.0 percentage points to Recall@1, 2.0 points to Recall@3, and 0.025 MRR relative to graph traversal OFF.

The two original result tables are reproduced below exactly as published in the main paper.

TABLE I

RETRIEVAL ABLATION AND GRAPH ISOLATION ON INSTITUTIONAL BENCHMARK ($N = 50$)

Configuration	Recall@1 (%)	Recall@3 (%)	MRR
Cumulative Component Ablation			
(1) Standard BM25 (Text only)	70.0 (35/50)	84.0 (42/50)	0.767
(2) + Document Title Weighting ($\alpha = 3.0$)	78.0 (39/50)	88.0 (44/50)	0.825
(3) + Intent Phrase Boosting ($\beta = 10.0$)	84.0 (42/50)	92.0 (46/50)	0.875
(4) + Title Inheritance (Graph Traversal OFF)	86.0 (43/50)	92.0 (46/50)	0.895
(5) **Full Proposed (Graph Traversal ON)**	**90.0** (45/50)	**94.0** (47/50)	**0.920**
Isolated Graph Retrieval Effect (Row 5 vs. Row 4)			
Standalone Graph Traversal Gain (Δ)	**+4.0%** (+2)	**+2.0%** (+1)	**+0.025**

As shown in Table I, each architectural component produces a distinct marginal improvement on the 50-scenario benchmark. Standard lexical BM25 achieves a baseline Recall@3 of 84.0%. Document title weighting increases Recall@1 from 70.0% to 78.0% and MRR to 0.825, while intent phrase boosting selectively elevates procedural circulars to 84.0% Recall@1.

Crucially, to address the limitation of purely cumulative ablation and strictly separate the contribution of graph traversal from static document title inheritance, configuration (4) serves as a dedicated isolation baseline: it retains title weighting ($\alpha = 3.0$), intent boosting ($\beta = 10.0$), and static parent title inheritance across attachment edges, but completely disables graph spreading activation ($S_{graph} = 0$). While static title inheritance successfully recovers orphaned attachment chunks (elevating Recall@1 to 86.0%), activating hierarchical graph traversal in configuration (5) yields an additional isolated improvement of +4.0% in Recall@1 (from 43 to 45 queries), +2.0% in Recall@3 (from 46 to 47 queries), and +0.025 in MRR (reaching 0.920).

This isolates the pure graph traversal contribution: personalized spreading activation uniquely resolves procedural inquiries requiring cross-departmental entity linking (e.g., connecting a student inquiry regarding tuition deferral or health insurance to the authoritative circular issued under Student Affairs or Academic Affairs jurisdiction via GOVERNS edges) that static title matching and keyword filtering cannot bridge.

5. Retrieval Results by Document Structure

1) Structural Breakdown: Narrative Text vs. Semi-Structured Tables: To address structural variability in institutional records, Table II decomposes retrieval performance across structural data modalities.

TABLE II
RETRIEVAL BREAKDOWN ACROSS DATA MODALITIES ($N = 50$)

Structural Modality	**Count** (N)	**Recall@1 (%)**	**Recall@3 (%)**	**MRR**
Narrative Institutional Text	26	92.3 (24/26)	96.2 (25/26)	0.942
Structured Tables and Forms	24	87.5 (21/24)	91.7 (22/24)	0.896
Overall Composite	**50**	**90.0** (45/50)	**94.0** (47/50)	**0.920**

While narrative documents exhibit higher recall (Recall@3 = 96.2%), structured tables and forms achieve 91.7% Recall@3 under our hierarchical graph model. The small performance gap highlights that multi-column OCR structures require robust graph-linked title inheritance to maintain competitive retrieval fidelity.

These table/form results concern structured records represented in the retrieval corpus; they do not constitute a separate image-based retrieval evaluation.

6. Answer Quality and Routing Results Reported in the Main Paper

1) Dialogue Routing Confusion Matrix: To verify that the query router discriminates effectively rather than behaving as a trivial majority classifier, Table III presents the contingency distributions for both the full 300-query benchmark and the balanced 60-query benchmark.

The original routing table is reproduced below exactly as published in the main paper.

TABLE III
DIALOGUE ROUTER CONFUSION MATRIX AND BALANCED METRICS

Evaluation Metric	**Full** ($N = 300$)	**Balanced** ($N = 60$)
True Positives (TP_{RAG})	267	30
False Negatives (FN_{RAG})	3	0
False Positives (FP_{Direct})	0	0
True Negatives (TN_{Direct})	30	30
Overall Accuracy (%)	99.00	100.00
Balanced Accuracy (%)	99.44	100.00
Specificity / TNR (%)	100.00	100.00
Macro F1-Score (%)	97.34	100.00

The zero false-positive rate (FP = 0) and 100% specificity across both cohorts directly confirm robust discrimination between administrative queries and open-domain chit-chat, disproving the always-positive hypothesis.

6. Answer Quality and Groundedness Results (continued)

2) Factual Grounding and Generation Quality Assessment: To validate generation fidelity without subjective bias, the 50-scenario institutional benchmark was evaluated across two distinct query subsets: (1) 40 In-Domain Policy Queries, each paired with verified ground-truth reference chunks and canonical key facts (e.g., exact hospital codes, fee amounts, deadlines, and submission forms); and (2) 10 Adversarial / Out-of-Domain Trap Queries querying non-existent majors (e.g., Aerospace Engineering) or ungrounded policies, specifically formulated to evaluate resistance against hallucination.

Two university academic advisors conducted a double-blind annotation of all 50 generated responses against official ground-truth university circulars. Annotators independently evaluated four explicit binary dimensions: (1) Factual Accuracy (1 if all stated numerical figures, office locations, and deadlines match canonical ground-truth circulars, 0 otherwise); (2) Context Groundedness / Faithfulness (1 if every generated proposition is strictly entailed by the retrieved context chunks); (3) Source Consistency (1 if the retrieved evidence chunks originate from the authoritative governing circular); and (4) Information Completeness (1 if all primary steps to complete the administrative procedure are conveyed). Inter-annotator agreement reached Cohen's $\kappa = 0.88$ for Factual Accuracy, $\kappa = 0.92$ for Groundedness, and $\kappa = 0.84$ for Information Completeness, indicating high concordance.

The original answer-quality table is reproduced below exactly as published in the main paper.

TABLE IV
FACTUAL ACCURACY AND GENERATION QUALITY METRICS ($N = 50$)

Quality Dimension	Measured Rate (%)
Factual Accuracy (Exact Evidence Match)	80.0 (40/50)
Groundedness (Context Entailment)	100.0 (50/50)
Source Consistency (Authoritative Authority)	94.0 (47/50)
Observed Hallucination Rate	0.0 (0/50)
Information Completeness	64.0 (32/50)
Acoustic Brevity Compliance ($\leq$ 300 chars)	100.0 (50/50)
Average Response Length (chars)	248.6

As documented in Table IV, no hallucinations were observed across the 50 evaluated responses (0.0% observed hallucination rate on the benchmark). The 20% non-exact factual cases (10/50) correspond precisely to the 10 adversarial/out-of-domain trap scenarios where no official university regulation exists. In all 10 instances, the agent reliably recognized context insufficiency and outputted a safe conversational refusal (e.g., 'Toi chua ro thong tin nay, ban vui long lien he phong Dao tao...' [I do not have sufficient information regarding this matter, please contact the Academic Affairs Office...]) with 100% compliance, successfully preventing the fabrication of deceptive regulations. Information completeness is 64.0% because spoken responses are intentionally bounded to concise actionable points (≤300 characters) for TTS delivery, filtering out verbose administrative legalese.

The qualitative examples table below is reproduced exactly as published. The paper presents two grounded, correct answers and one safe refusal for an out-of-domain query; no separate explanatory paragraph follows Table V in the main paper.

TABLE V
REPRESENTATIVE QUALITATIVE EVALUATION EXAMPLES ON THE GOLDEN INSTITUTIONAL BENCHMARK

Student Query (u_t)	**Ground-Truth Policy Target**	**HIRA Generated Response**	**Governing Source Reference**	**Evaluation Outcome**
"Benh vien Nhan dan Gia Dinh co so 2 co ma co so BHYT la bao nhieu va dia chi o dau?" [What is the BHYT code and address of Gia Dinh Hospital Facility 2?]	Initial healthcare clinic listing: Code `79-001`, address 125 Le Loi, Ward 3, Go Vap District.	*"Ma co so KCB BHYT ban dau cua Benh vien Nhan dan Gia Dinh (Co so 2) la 79-001, dia chi tai 125 Le Loi, Phuong 3, Quan Go Vap, TP.HCM."* (128 chars)	Circular on Initial Healthcare Facilities 2025–2026, Table Appendix `chunk_007`.	Acc: 1 Grounded: 1 Hallucination: 0
"Sinh vien nop don xin gia han hoc phi hoc ky 2 o phong ban nao va han chot la khi nao?" [Which office receives tuition extension forms and what is the deadline?]	Tuition deferral jurisdiction: Academic Affairs Office (*Phong Dao tao*), deadline February 12, 2026.	*"Don xin gia han dong hoc phi hoc ky 2 duoc tiep nhan boi Phong Dao tao (khong nop tai Phong CTSV). Han chot nop don la ngay 12/02/2026."* (136 chars)	Decision on Tuition Extension, Academic Affairs Notice `chunk_003`.	Acc: 1 Grounded: 1 Hallucination: 0
"Hoc phi nganh Ky thuat Hang khong Vu tru tai HUFLIT nam 2026 la bao nhieu mot tin chi?" [What is the tuition per credit for Aerospace Engineering at HUFLIT in 2026?]	Adversarial trap query: Program is not offered at HUFLIT (Unsupported institutional domain).	*"Hien tai nha truong khong dao tao nganh Ky thuat Hang khong Vu tru. Ban vui long lien he Phong Dao tao de duoc huong dan danh muc nganh tuyen sinh chinh thuc."* (158 chars)	Adversarial Out-of-Domain Probe (No matching circular).	Acc: 0 (Safe) Grounded: 1 Hallucination: 0 Refusal: 1

Table V illustrates two supported answers (a healthcare code/address and a tuition-deferral office/deadline) and one safe refusal for a program outside the university's offerings. The Source column labels such as chunk_007 and chunk_003 are evidence-passage references retained from the original table; they are not user or participant IDs.

7. Stored Automatic Counseling Evaluation

Summary of the 300-row Qwen3.5-4B-Instruct-labeled evaluation record.

Measure	Result	Denominator
Correct verdict	198 (66.00%)	300
Partially correct verdict	37 (12.33%)	300
Incorrect verdict	65 (21.67%)	300
Mean stored score (0–1)	0.7217 (72.17%)	300
Designated source ID returned	166 (55.33%)	300
Designated source ID not returned	134 (44.67%)	300

Table 4. Stored automatic answer judgments and designated-source document hits.

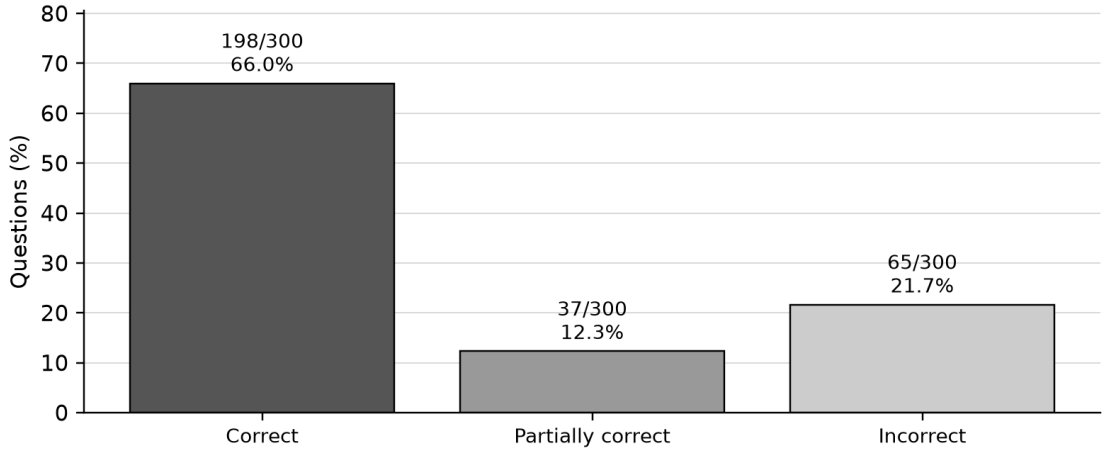


Figure 5. Distribution of stored automatic answer verdicts.

The evaluator marked 66.00% of answers correct, 12.33% partially correct, and 21.67% incorrect; partial answers contribute 0.5 to the reported mean score of 72.17%. The designated source ID appeared in 55.33% of returned result lists. This source-presence check and the answer-judgment score measure different aspects of the run.

This is a separate 300-row automatic evaluation record, not the 50-query human-assessed benchmark in Section 6. These are automatic LM Studio judge outputs, not human-verified QA metrics. The row-level CSV does not independently record the evaluator model ID. The source-hit rate is not Recall@3. No human-verified completeness, groundedness, hallucination, or source-consistency rates are reported for this 300-row record.